

Diego Rodríguez Langevin

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EDUCATION

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|-----------|---|
| 2022-2025 | M.Phil., New York University (Neural Science). |
| 2017-2022 | Sc.B., Brown University (Neuroscience, honors)
A.B., (Philosophical Inquiry Through Creative Forms, honors)
A.B., (Computer Science). |
| 2019 | Visiting Student, Syracuse University in Florence, Italy (Studio Arts). |

RESEARCH EXPERIENCE

New York University, Center for Neural Science

2023–2025 | *Co-advisors: Eric Klann and André Fenton*

- Investigated the effects of ketamine on hippocampal neural oscillations and de novo protein synthesis through a joint systems and molecular neuroscience project.
- Used **FUNCAT metabolic labeling, confocal microscopy, ImageJ, and Prism** to quantify nascent protein synthesis in mouse hippocampus following ketamine treatment.
- Analyzed single-unit and population-level activity from **Neuropixels recordings** during ketamine treatment.
- Developed **Python** analysis notebooks and managed high-performance computing environments for large-scale neural recording and proteomic datasets.

NYU Langone Health, Neuroscience Institute

2023 | *Advisor: Richard Tsien*

- Investigated how schizophrenia-associated non-coding variants in **CACNA1C** may alter neuronal function through alternative RNA splicing.
- Identified candidate regulatory regions using the **UCSC Genome Browser, RBPmap, and Ensembl**, and analyzed human iPSC-derived neurons carrying CRISPR-mediated deletions.
- Performed **real-time PCR** on cDNA from human iPSC-derived neurons to assess effects of candidate variants on RNA splicing.

Brown University, Department of Neuroscience

2019–2022 | *Advisor: Alexander Fleischmann*

- Investigated the genetic identity and connectivity of cortical neurons engaged in associative learning using mouse olfactory models.
- Used **doxycycline-controlled Fos tagging, iDISCO tissue clearing, and light-sheet microscopy** to map connectivity of behaviorally activated neurons.

- Performed **single-cell transcriptomic analysis** using R, Seurat, Bioconductor, and Harmony to identify the genetic identity of Fos-tagged cells.
- Analyzed spatial transcriptomic data using **R and ImageJ** to identify regions of interest based on olfactory receptor gene expression.

GRANTS

2024-2025 National Institute of Mental Health’s Training Systems and Integrative Neuroscience Grant (T32 MH019524). “The Effect of Ketamine on Protein Synthesis and Neural Oscillations in the Hippocampus.”

AWARDS AND FELLOWSHIPS

2022-2024 McCracken Graduate Fellowship (Center for Neural Science, New York University).

2022 Sigma Xi (Brown University).

2022 Class Marshal (Brown University).

2020 LINK Research Award in Neuroscience (Brown University, summer term).

2020 Undergraduate Teaching and Research Award (with Prof. Alexander Fleischmann) in Neuroscience (Brown University, spring term).

RESEARCH PRESENTATIONS

2023 Speaker, “The Effects of Schizophrenia Associated Variations on the Alternative Splicing of *CACNA1C*,” First Year Talks, Neuroscience Institute, New York University, New York, NY (December 6).

2022 Poster Presentation, “Fos-Tagging of Olfactory Engrams: Using Single-Nuclei RNA Sequencing and Whole-Brain Clearing Histology to Characterize and Map Neurons Involved in Olfactory Associative Learning and Memory,” Honors Poster Presentations, Department of Neuroscience, Brown University, Providence, RI (May 3).

TEACHING

New York University, Graduate Teaching Assistant

NEURL-GA 2201: *Cellular Neuroscience*, Fall 2024.

NEURL-UA 100: *Introduction to Neural Science*, Spring 2024.

Brown University, Undergraduate Teaching Assistant

CSCI 1430: *Computer Vision*, Spring 2022 (Socially Responsible Computing TA).

NEUR 0010: *Introduction to Neuroscience*, Fall 2020 (online), Summer 2021.